

Matrices: Many Dot Products at Once

The table of numbers behind every layer · video 2:02

This lesson has one big idea: a **matrix** is a table of numbers that does **many dot products at once**. Matrix times vector dots every row with the vector; matrix times matrix dots every row with every column. That is how one layer of an LLM transforms every token in a single step.

How to use this guide: master one concept before moving to the next.

1. Watch the whole video once, without stopping.
2. Then take the concepts **in order**. For each one: read the short explanation, look at the picture from the video (▶ is the time in the video), and answer every question in its *Check* without looking anything up.
3. Check your answers in the **Answers** section at the end (page 10).
4. **Pass mark: every question in the check correct.** If you missed one, rewatch that concept's video segment, reread the explanation, and redo only the questions you missed. Tick the concept off below, then move on.

It is strongly suggested that you **master each concept before starting the next one**. Each concept builds on the previous ones.

Concepts in this lesson

CONCEPT	VIDEO SEGMENT	QUESTIONS	MASTERED
1. A matrix is a table of numbers	▶ 0:08-0:25	4	☐
2. Matrix times vector	▶ 0:25-0:36	4	☐
3. A matrix transforms space	▶ 0:36-0:47	4	☐
4. The shapes must line up	▶ 0:47-0:58	4	☐
5. Matrix times matrix: every token at once	▶ 0:58-1:24	4	☐
6. The transpose	▶ 1:24-1:36	4	☐
7. Matrices in NumPy	▶ 1:36-1:59	3	☐

Before you start: this builds on F01 (Vectors) and F02 (The Dot Product): every matrix product here is a set of dot products. Matrices power main episodes 5, 6 and 7 (Attention), 8 (The MLP) and 10 (From Vectors Back to Words). Concept 7 reads a few lines of NumPy.

CONCEPT 1 OF 7

A matrix is a table of numbers

▶ Video 0:08–0:25

Language models compute billions of dot products; writing them one by one would be hopeless.

Matrices let us do many of them at once. A matrix is a **table of numbers**, with **rows** and **columns**:

$$M = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \end{pmatrix}$$

This M has 2 rows and 3 columns, so we call it a **2 × 3 matrix** (“two by three”). The shape is always **rows × columns**, in that order. Each row is a vector: row 1 of M is [1, 0, 2].

$$M = \begin{matrix} & \text{col 1} & \text{col 2} & \text{col 3} \\ \begin{matrix} \text{row 1} \\ \text{row 2} \end{matrix} & \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \end{bmatrix} \end{matrix}$$

2 × 3 matrix (rows × columns)

▶ 0:25 M has 2 rows and 3 columns: a 2 × 3 matrix. Shapes are always rows × columns.

KEY IDEA

A matrix is a **table of numbers**. Its shape is **rows × columns**: M is 2 × 3.

Check 1: A matrix is a table of numbers

Answer all 4 questions before looking at the answers. Pass mark: all correct.

1.1 CALCULATION

$A = \begin{pmatrix} 2 & 7 & 1 & 0 \\ 5 & 3 & 8 & 4 \end{pmatrix}$. What is its shape, and which number is in row 2, column 3?

1.2 TRUE OR FALSE

“A 2 × 3 matrix and a 3 × 2 matrix have the same shape, since both hold 6 numbers.”

True ☐ False ☐

1.3 SHORT ANSWER

Write down row 2 and column 3 of the matrix M above.

1.4 CALCULATION

GPT-2's vectors have 768 numbers. How many numbers does a 768 × 768 matrix hold?

CONCEPT 2 OF 7

Matrix times vector

▶ Video 0:25–0:36

To multiply a matrix by a vector, take the **dot product of each row with the vector**. Each row gives one number, so the result is a new vector, with **one entry per row**.

$$M \times = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 7 \\ -1 \end{pmatrix}$$

Row 1: $1 \cdot 3 + 0 \cdot 1 + 2 \cdot 2 = 7$. Row 2: $0 \cdot 3 + 1 \cdot 1 + (-1) \cdot 2 = -1$. So $M \times = [7, -1]$.

$$\begin{array}{ccc} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \end{bmatrix} & \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix} & = \begin{bmatrix} 7 \\ -1 \end{bmatrix} \\ M & x & M \times \\ 1 \cdot 3 + 0 \cdot 1 + 2 \cdot 2 = 7 & & \\ 0 \cdot 3 + 1 \cdot 1 + (-1) \cdot 2 = -1 & & \\ \text{one entry per row} & & \end{array}$$

▶ 0:35 Each row of M dotted with x gives one number: 7 from row 1, -1 from row 2.

KEY IDEA

$M \times$ = **one dot product per row** of M. The result has one entry per row.

Check 2: Matrix times vector

Answer all 4 questions before looking at the answers. Pass mark: all correct.

2.1 CALCULATION

Compute $\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 4 \\ 5 \end{pmatrix}$.

2.2 CALCULATION

Compute $M y$ for the video's M and $y = [1, 1, 1]$.

2.3 MULTIPLE CHOICE

What does $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ do to any vector $[p, q]$?

- A. Leaves it unchanged: $[p, q]$
- B. Swaps the two numbers: $[q, p]$
- C. Doubles it: $[2p, 2q]$
- D. Flips it: $[-p, -q]$

2.4 TRUE OR FALSE

"Each entry of $M \times$ uses all the numbers of x, but only one row of M."

True ☐ False ☐

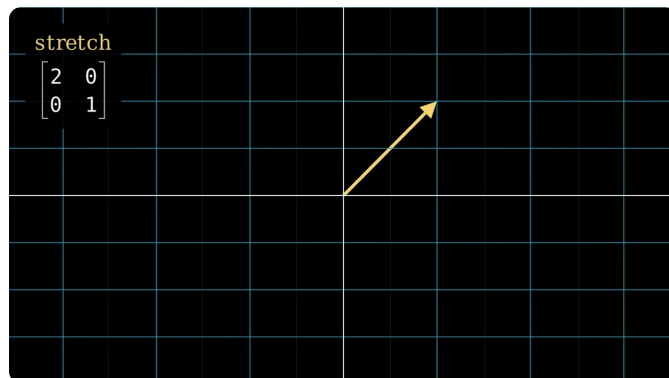
CONCEPT 3 OF 7

A matrix transforms space

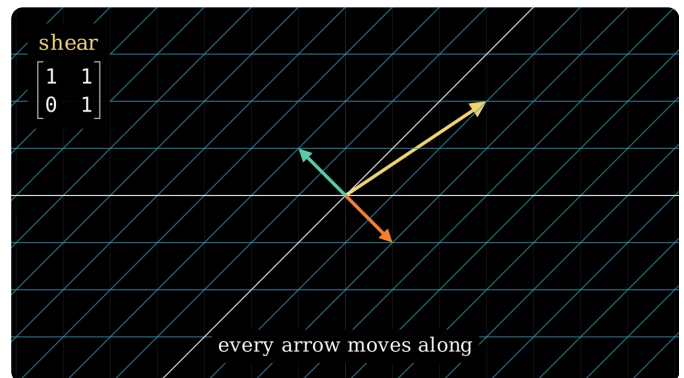
▶ Video 0:36–0:47

So a matrix **turns one vector into another**. In two dimensions you can watch it happen: apply the matrix to every point and the whole plane moves. A matrix can **stretch** space, **rotate** it, or **shear** it, and every arrow moves along. The video's three matrices, applied to the yellow arrow $[1, 2]$:

$$\text{stretch} \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \rightarrow [2, 2] \quad \cdot \quad \text{rotate } 90^\circ \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \rightarrow [-2, 1] \quad \cdot \quad \text{shear} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \rightarrow [3, 2]$$



▶ 0:43 Stretch: every x-coordinate doubles; y stays the same.



▶ 0:47 Shear: space slides sideways, and every arrow moves along.

KEY IDEA

A matrix is a **function on vectors**: vector in, vector out. It moves every arrow in the same systematic way, such as a stretch, a rotation or a shear.

Check 3: A matrix transforms space

Answer all 4 questions before looking at the answers. Pass mark: all correct.

3.1 CALCULATION

Apply the rotate matrix $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ to $[3, 1]$.

.....

3.2 CALCULATION

Apply the stretch matrix $\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ to $[-1, 3]$.

.....

3.3 CALCULATION

Write down a 2×2 matrix that makes every arrow 3 times as long, without changing its direction.

.....

3.4 TRUE OR FALSE

"Applying the rotate matrix four times brings every arrow back to where it started."

True ☐ False ☐

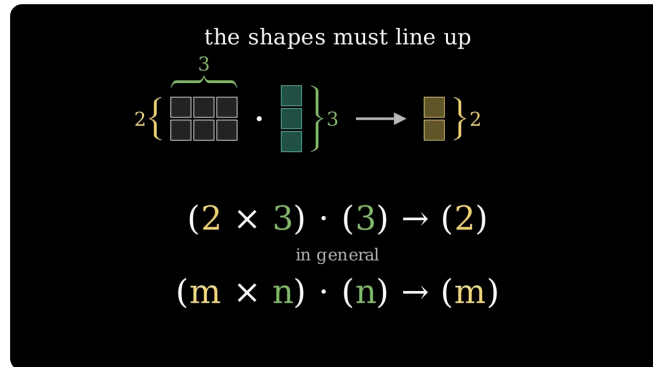
CONCEPT 4 OF 7

The shapes must line up

▶ Video 0:47-0:58

The **shapes must line up**. Each row is dotted with the vector, so the vector needs exactly as many numbers as the matrix has **columns**. A 2×3 matrix needs a vector with 3 numbers, and gives back 2, one per row.

In general, an $m \times n$ matrix turns n numbers into m numbers: $(m \times n) \cdot (n) \rightarrow (m)$.



▶ 0:57 A 2×3 matrix needs 3 numbers and gives back 2. In general, $(m \times n) \cdot (n) \rightarrow (m)$.

KEY IDEA

Columns = how many numbers go in; **rows** = how many come out.

Check 4: The shapes must line up

Answer all 4 questions before looking at the answers. Pass mark: all correct.

4.1 CALCULATION

A 4×6 matrix multiplies a vector. How many numbers must the vector have, and how many come out?

4.2 MULTIPLE CHOICE

You want a matrix that turns a 768-number vector into a 3,072-number vector. What shape must it have?

- A. $768 \times 3,072$
- B. $3,072 \times 768$
- C. 768×768
- D. $3,072 \times 3,072$

4.3 TRUE OR FALSE

"A 3×2 matrix can multiply the vector $[3, 1, 2]$."

True ☐ False ☐

4.4 CALCULATION

A vector of 5 numbers is multiplied by A (shape 8×5), and the result by B (shape 2×8). How many numbers come out at the end?

CONCEPT 5 OF 7

Matrix times matrix: every token at once

▶ Video 0:58-1:24

Multiplying two matrices is the same idea, repeated: **every row of the first, dotted with every column of the second**. The inner sizes must match and the outer sizes give the result: $(5 \times 4) \cdot (4 \times 4) \rightarrow 5 \times 4$. In general, $(m \times n) \cdot (n \times p) \rightarrow (m \times p)$.

That's exactly how LLMs use it: stack the tokens' vectors as the **rows** of X , multiply by one **weight matrix** W , and every token is transformed in a single step. Row i of $X \cdot W$ belongs to token i .

row 1 · column 1 = $1 \cdot 1 + (-2) \cdot (-2) + 1 \cdot (-1) + 1 \cdot 2 = 6$

illustrative values

▶ 1:06 Row 1 · column 1 = 6 (illustrative).

every token, one multiplication

▶ 1:23 All five tokens in one product.

KEY IDEA

$A \cdot B$ = **every row of A dotted with every column of B**. With tokens as rows, $X \cdot W$ transforms them all at once.

Check 5: Matrix times matrix: every token at once

Answer all 4 questions before looking at the answers. Pass mark: all correct.

5.1 CALCULATION

Compute $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \cdot \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$.

5.2 CALCULATION

In the video, row 2 of X is $[2, -1, 0, 2]$ and column 1 of W is $[1, -2, -1, 2]$. Which entry of $X \cdot W$ do they give, and what is it?

5.3 MULTIPLE CHOICE

Which product is **not** allowed?

- A. $(5 \times 4) \cdot (4 \times 4)$
- B. $(2 \times 3) \cdot (3 \times 1)$
- C. $(4 \times 5) \cdot (4 \times 5)$
- D. $(1 \times 768) \cdot (768 \times 768)$

5.4 CALCULATION

10 tokens of 768 numbers form X ; W is 768×768 . (a) Shape of $X \cdot W$? (b) How many dot products? (c) Which row holds the transformed 4th token?

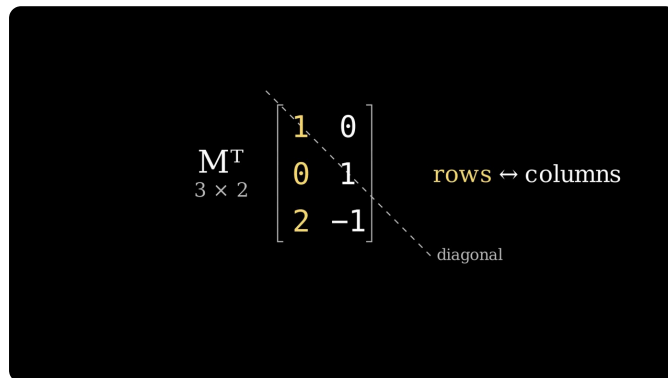
CONCEPT 6 OF 7

The transpose

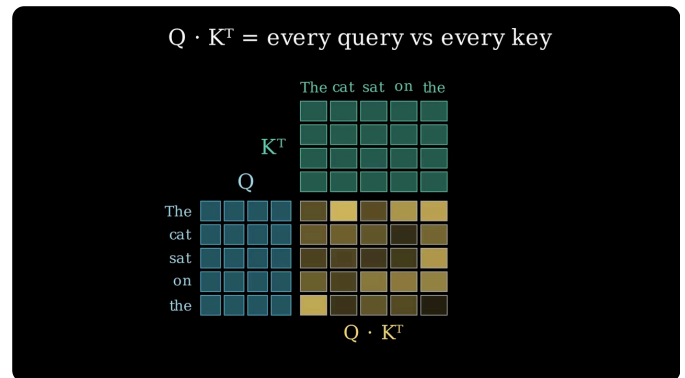
▶ Video 1:24-1:36

One more operation: the **transpose**, written M^T . It flips a matrix over its **diagonal**, so **rows become columns**: row 1 of M , $[1, 0, 2]$, becomes column 1 of M^T , and the 2×3 M becomes 3×2 .

Why it matters: in attention, Q holds a *query* vector per token and K a *key* vector per token, both as rows. To dot every query with every key, the keys must become columns: $Q \cdot K^T$. With 5 tokens of 4 numbers each, $(5 \times 4) \cdot (4 \times 5) \rightarrow 5 \times 5$: one score for every pair of tokens.



▶ 1:30 M flipped over its diagonal: M^T is 3×2 .



▶ 1:36 $Q \cdot K^T$: every query against every key.

KEY IDEA

The transpose **swaps rows and columns** ($m \times n \rightarrow n \times m$). $Q \cdot K^T$ compares every query with every key in one product.

Check 6: The transpose

Answer all 4 questions before looking at the answers. Pass mark: all correct.

6.1 SHORT ANSWER

Transpose $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$. What shape is the result?

6.2 TRUE OR FALSE

"Transposing a matrix twice gives back the original matrix."

True ☐ False ☐

6.3 MULTIPLE CHOICE

Q and K both have shape 5×4 (5 tokens, 4 numbers each). Why can't we just compute $Q \cdot K$?

- A. The inner sizes don't match (4 and 5)
- B. Only square matrices can be multiplied
- C. $Q \cdot K$ gives the same result as $Q \cdot K^T$ anyway
- D. Two matrices of the same shape can never be multiplied

6.4 CALCULATION

A prompt has 12 tokens, with query and key vectors of 64 numbers. What shape is $Q \cdot K^T$, and how many scores does it hold?

CONCEPT 7 OF 7

Matrices in NumPy

▶ Video 1:36-1:59

In NumPy, the @ sign multiplies matrices too (matrix × vector and matrix × matrix), .T gives the transpose, and .shape tells you the shape. Always check the shapes.

```
1 M = np.array([[1, 0, 2],
2               [0, 1, -1]])      # 2 x 3
3 x = np.array([3, 1, 2])
4 M @ x
5 X = np.random.randn(5, 4)      # 5 tokens, 4 numbers each
6 W = np.random.randn(4, 4)
7 (X @ W).shape                  # (5, 4)
8 M.T.shape                      # (3, 2)
```

▶ 1:43 The code from the video: @ multiplies matrices too, .T transposes, and .shape lets you check shapes.

```
M = np.array([[1, 0, 2],
               [0, 1, -1]])      # 2 x 3
x = np.array([3, 1, 2])
M @ x
X = np.random.randn(5, 4)      # 5 tokens, 4 numbers each
W = np.random.randn(4, 4)
(X @ W).shape                  # (5, 4)
M.T.shape                      # (3, 2)
```

Matrices power almost every step from episode 5 to episode 10: queries, keys and values, attention scores, the MLP, and the final logits.

KEY IDEA

@ multiplies, .T transposes, .shape checks: $(m \times n) @ (n \times p) \rightarrow (m \times p)$.

Check 7: Matrices in NumPy

Answer all 3 questions before looking at the answers. Pass mark: all correct.

7.1 SHORT ANSWER

With M from the code above, what is `(M.T @ M).shape`?

7.2 MULTIPLE CHOICE

What happens when you run `M @ M`?

- A. You get an array of shape (2, 3)
- B. A ValueError: the shapes don't line up
- C. Every number in M is squared
- D. You get an array of shape (2, 2)

7.3 HANDS-ON CODE

Try it yourself. (a) Complete `matvec` so it gives the same result as `M @ x`, using one dot product per row. (b) `X` holds 4 tokens as rows. Why does `X @ M` fail, while `X @ M.T` works? What shape does `X @ M.T` have? (c) Compare the first row of `X @ M.T` with `matvec(M, X[0])`. What do you notice?

```
import numpy as np

M = np.array([[1, 0, 2],
              [0, 1, -1]])
x = np.array([3, 1, 2])

def matvec(M, x):
    out = []
    for row in M:
        out.append(...)      # (a) your code here
    return np.array(out)

print(matvec(M, x), M @ x)

X = np.array([[1, 2, 0],      # 4 tokens as rows,
              [0, 1, 1],      # 3 numbers each
              [2, 0, 1],
              [1, 1, 1]])

print(X @ M.T)
print(matvec(M, X[0]))
```

ANSWERS

Answers and explanations

Mark each answer right or wrong. If a concept has any wrong answer, rewatch its segment (▶) and redo the questions you missed before moving on.

1. A matrix is a table of numbers · rewatch ▶ 0:08-0:25

1.1 2×4 ; the number 8. 2 rows, 4 columns. Row 2 is [5, 3, 8, 4], and its third entry is 8.

1.2 False. Shape is rows $\times$ columns, and the order matters: 2 rows of 3 is not 3 rows of 2.

1.3 Row 2 = [0, 1, -1]; column 3 = [2, -1]. A row goes across (3 numbers in a 2×3 matrix); a column goes down (2 numbers).

1.4 589,824. 768 rows $\times$ 768 columns = 589,824. Weight matrices get big fast.

2. Matrix times vector · rewatch ▶ 0:25-0:36

2.1 [13, 15]. Row 1: $2 \cdot 4 + 1 \cdot 5 = 13$. Row 2: $0 \cdot 4 + 3 \cdot 5 = 15$.

2.2 [3, 0]. Dotting with [1, 1, 1] just adds up each row: $1 + 0 + 2 = 3$ and $0 + 1 - 1 = 0$.

2.3 B. Row 1 [0, 1] picks out q; row 2 [1, 0] picks out p.

2.4 True. Entry i is row i of M dotted with the whole of x.

3. A matrix transforms space · rewatch ▶ 0:36-0:47

3.1 [-1, 3]. Row 1: $0 \cdot 3 + (-1) \cdot 1 = -1$. Row 2: $1 \cdot 3 + 0 \cdot 1 = 3$. As F02 showed, [-1, 3] is at right angles to [3, 1].

3.2 [-2, 3]. The first number doubles, the second stays the same.

3.3 Rows [3, 0] and [0, 3]. It sends [p, q] to [3p, 3q] = $3 \times [p, q]$. The video's stretch matrix changes only the first number, so it stretches sideways only.

3.4 True. $4 \times 90^\circ = 360^\circ$, a full turn.

4. The shapes must line up · rewatch ▶ 0:47-0:58

4.1 6 in, 4 out. Columns (6) = input size; rows (4) = output size.

4.2 B. $m \times n$ turns n numbers into m: 3,072 rows (one per output), 768 columns (one per input).

4.3 False. A 3×2 matrix has 2 columns, so it needs 2 numbers. The video's 2×3 M is the one that takes [3, 1, 2].

4.4 2. A turns 5 numbers into 8, then B turns 8 into 2. The shapes line up at each step.

5. Matrix times matrix: every token at once · rewatch ▶ 0:58-1:24

5.1 Rows [19, 22] and [43, 50]. Top-left: row 1 $\cdot$ column 1 = $1 \cdot 5 + 2 \cdot 7 = 19$. Likewise $1 \cdot 6 + 2 \cdot 8 = 22$, $3 \cdot 5 + 4 \cdot 7 = 43$, $3 \cdot 6 + 4 \cdot 8 = 50$.

5.2 Row 2, column 1: 8. $2 \cdot 1 + (-1) \cdot (-2) + 0 \cdot (-1) + 2 \cdot 2 = 2 + 2 + 0 + 4 = 8$, as in the finished result in the video.

5.3 C. The inner sizes are 5 and 4, which don't match. The others give 5×4 , 2×1 and 1×768 .

5.4 (a) 10×768 (b) 7,680 (c) row 4. $(10 \times 768) \cdot (768 \times 768) \rightarrow 10 \times 768$; one dot product per entry, $10 \times 768 = 7,680$; row i of the result belongs to token i.

6. The transpose · rewatch ► 1:24-1:36

6.1 Rows [1, 4], [2, 5], [3, 6]; shape 3×2 . Row 1 [1, 2, 3] becomes column 1, row 2 [4, 5, 6] becomes column 2.

6.2 True. Flipping over the diagonal twice puts every number back where it was.

6.3 A. K^T is 4×5 , so $(5 \times 4) \cdot (4 \times 5)$ lines up and gives 5×5 . B and D are false: only the inner sizes matter.

6.4 12×12 ; 144 scores. $(12 \times 64) \cdot (64 \times 12) \rightarrow 12 \times 12$: one score for every (query, key) pair.

7. Matrices in NumPy · rewatch ► 1:36-1:59

7.1 (3, 3). M.T is 3×2 and M is 2×3 : $(3 \times 2) @ (2 \times 3) \rightarrow 3 \times 3$.

7.2 B. $(2 \times 3) @ (2 \times 3)$: the inner sizes 3 and 2 don't match. Squaring every number (C) would be $M * M$.

7.3 (a) [7 -1] [7 -1] · (b) shape (4, 2) · (c) both are [1 2] (a) `out.append(row @ x)`. (b) $X @ M$ is $(4 \times 3) @ (2 \times 3)$: the inner sizes 3 and 2 don't match, so NumPy raises a `ValueError`. M.T is 3×2 , so $(4 \times 3) @ (3 \times 2) \rightarrow (4, 2)$: two numbers per token. (c) Both are [1, 2]. Each row of $X @ M.T$ is M applied to that token, so one product applies M to all four tokens at once.